Home Exam BIOS13 Modelling Biological Systems

Dept. of Biology, Lund University, 2025-2026

Deadline: Tuesday, Jan 13th, 2025 at 16:00

Instructions (written exam)

Solve the problems on your own, without discussing with your classmates. Needless to say, working in groups or copying text from elsewhere will be cheating and thus render repercussions. Cross-referencing of answers will be done within the class but also to exam answers from previous years and the internet. Email me questions if you need clarifications or have other issues. I will respond to the whole group.

The exam is set up as four separate assignments, corresponding to the four (main) questions found below, numbered 1-4. Prepare your answers as one main document per question (in total 4). Use Word or pdf format and all text should be written in a text editor. Scanned or photographed hand-written equations or illustrations are allowed, *as long as they are easy to read and properly embedded in the total answer*.

Please paste the required R code in the answer document, with the code in the word/pdf-document it is easier for me to comment on the code. In addition, make sure to provide the code as separate R-files (you can submit several files to the same assignment). Do make sure it is obvious which code belongs to which (sub-)question! Also, do make sure that I can run the scripts, code that does not run will not be considered! Test them yourselves by clearing the workspace and using source:

```
> rm(list=ls())
> source('my_script_1.R')
```

Any code that solves the task at hand is fine with me. It is perfectly allowed to use modified scripts from exercises or lectures. *However*, make sure to remove bits of code that do not contribute to the solution of the problem at hand. Otherwise, I may get the impression you do not know what you are doing and will grade accordingly.

Present the necessary steps of all calculations, at least briefly. Even an incorrect answer can give you points if the equations were put up correctly.

If a problem depends on the answer of a previous problem, and you failed to solve the previous problem, at least describe how you would proceed *if* you had the answer.

Submit your answers at the very latest on **Tuesday, Jan 13th, 2026 at 16:00**

Instructions (oral exam)

Upload one PowerPoint presentation per question 2-4 and prepare to present it in a 10-minute oral presentation. Be prepared to present all three presentations (I will choose which one(s)) at the oral exam session.

The presentation should clearly describe the following:

1. The overarching scope of the problem (1-2 sentences). Put the exam problem in a broad context.
2. The specific scope of the problem (1-2 sentences). Why is these problems of particular interest for science?
3. Specific questions (essentially the questions posed in the exam)
4. General description of how you answered the questions (e.g. "*I solved question a. by solving equation 1 mathematically by rearranging…*" or "*I solved question b. by coding a simulation model that includes…* ").
5. Specific description of how you answered the questions (e.g. "*Specifically I used the chain rule and I implemented my simulation using a nested for-loop*"). Show equations and code if needed but illustrations for code structure is also good.
6. Clear results in equations and plots (mind notation, panel titles, axes, scales, etc)
7. Interpretation of results given your scope

Note: Put a **strong** emphasis on points 4 and 5 above as this is what is evaluated. At least 7 minutes should be devoted to these points. Also make sure that you stick to 10 minutes, I will interrupt you when time is up.

GOOD LUCK!

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1. Adatpation and Genetic Algorithms (7p)

We aim to optimize the function g(x,y) = (1 - x)^2 + 100(y - x^2)^2 + 2*sin(3x), under domain -3 ≤ x,y ≤ 3.

Each candidate solution is represented as a vector (x,y).

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a) Describe what kinds of optimization problems GAs are particularly suited for. Is the problem above suitable for a GA, if so why? (1p)

b) Explain how selection, recombination, mutation, and replacement relate to biological evolution and GA alike. (1p)

c) Write an R function that takes two parent chromosomes and returns an offspring using one-point crossover. (1p)

d) Modify your function in (c) to include mutation with probability 0.1 and truncated to the valid range. (1p)

e) Write a full GA simulation in R using population size n=50, for at least 50 generations, with ranking selection, recombination, mutation, and a plot of mean and best fitness over time. (2p)

f) The crossover operator used above only swaps gene positions between parents (i.e., offspring genes always come directly from one parent or the other), and the mutation probability is relatively low. Explain why this may limit the GA's ability to find the global optimum of the function, even after many generations. Provide a clear conceptual explanation (1p)

2. Density dependence of an unstructured population (5p)

Consider a population that grows according to

$$\frac{dn}{dt} = rn\left(1 - \left(\frac{n}{K}\right)^{\theta}\right),$$

where r, K and θ are positive constants.

a) What is the (non-trivial) equilibrium population size? (1p)

b) Show that it is a stable equilibrium. (1p)

c) Write a script in R that plots, in a single diagram, the function $f(n) = rn\left(1 - \left(\frac{n}{K}\right)^{\theta}\right)$ vs. n, for a suite of θ-values, ranging from 0.2 to 3. (1p)

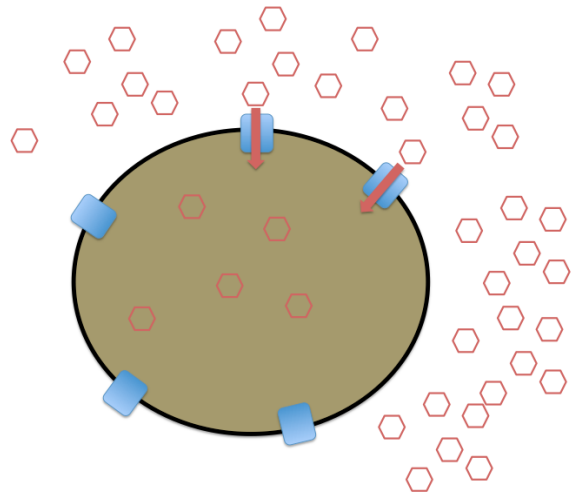
d) Write another script in R that plots, in a single diagram, the *per capita growth rate* $\frac{f(n)}{n}$ vs. n for a suite of θ-values, ranging from 0.2 to 3. (1p)

e) The plot in d) is a good illustration of the *density dependence*, i.e. how population density (size) affects population growth. Can you describe, in biological terms, the effect of θ on this density dependence? Put another way, what ecological

circumstance would render a low or high θ? An (imaginary) example or two could be useful. (1p)

3. Nutrient uptake of a cell (6p)

Cells consume nutrients, such as sugars, but the nutrients first have to pass through the cell membrane from the surrounding fluid. The transport of nutrients through the cell membrane depends on the surrounding nutrient concentration, but also on the number of receptors (transport proteins) in the membrane. It can be shown that the nutrient concentration c in the surrounding fluid follows (approximately)



$$\frac{dc}{dt} = -\frac{K_{max}c}{k_n + c}n$$

where n is the density of cells (per unit volume), $K_{\max}$ is the maximal uptake rate for a single cell and k_n is a half-saturation constant. Both $K_{\max}$ and k_n depend on the number of receptors per cell and their efficiency. The equation above is sometimes called *Michaelis-Menten kinetics* (Edelstein-Keshet, 1988, *Mathematical Models in Biology*).

a) The differential equation above describes the decrease of c in the surrounding fluid, which corresponds to the total uptake of all cells. Show (somehow) that the maximal uptake per cell is K_{max} as c goes to infinity? (1p)

b) Confirm (mathematically) that $c = 0$ is an equilibrium, and that it is stable (1p)

c) Write an R script that solves the differential equation above and plots the result (c as a function of time). All model parameters should be positive, but otherwise arbitrary. (1p)

Now assume we have chemostat conditions, i.e. that there is a constant inflow I of nutrients and an outflow μc proportional to the current concentration.

d) Write the new equation for the nutrient dynamics! (1p)

e) Adjust the R-script according to the new model. (1p)

f) Is there a new equilibrium? Is it stable? (Use either simulations in R or your own calculations for this) (1p)

4. Demographic stochasticity (3p)

Small populations are vulnerable to extinction simply because they are small. The fate of single individuals may have large effect on the growth of the population. A few failed breeding attempts or fatal infections can cause a dramatic population decline. For large populations these effects tend to average out, such that the population as a whole has more predictable growth. This source of uncertainty is called *demographic stochasticity*. A model of population growth taking demographic stochasticity is described om words below.

Assume a (asexual) population of N individuals. Each year, each individual produces a few offspring and then dies. The offspring constitute the next generation, and so on. The number of offspring produced by a single individual follow a *Poisson* distribution with a mean of 1. Thus, each individual replaces itself *on average*, but there may be considerable variation between individuals. Some are lucky and get many offspring, some get none. Total population size N will in this manner vary over time and may even reach extinction, $N = 0$.

In *R*, the function `rpois(N,lambda)` generates N random numbers with mean *lambda*.

a) Write a *function* in *R* that takes parameters *N0* and *tmax* as input and returns the number of individuals in the population after *tmax* generations, starting with *N0* individuals. (1p)

b) Write another *function* that takes *N0* and *tmax* as input parameters and calls the function from a) repeatedly (1000 times) to calculate the risk of extinction within *tmax* generations, starting with *N0* individuals. (1p)

c) Finally, write a script that uses the function from b) to plot the risk of extinction within 100 generations as a function of *N0*, with *N0* taking the values 10, 20, 30, … 100. (1p)